Basic Details of the Team and Problem Statement

Official SIH idea-submission cover • blind review: no institute details

RaahSetu

AI-Based Smart Logistics & Accessibility Intelligence Platform for the Northeast

Standard apps show the fastest route.
RaahSetu finds the safest feasible route — avoiding landslide, flood and accident-blackspot exposure — and explains the trade-off.

Software • Smart Automation / Logistics

Ministry / Organization:

Ministry of Development of North Eastern Region

Problem Statement No.:

SIH26002

Problem Statement Title:

AI-Based Smart Logistics & Accessibility Intelligence Platform

Idea Title:

RaahSetu — Risk-Aware A* Routing + Explainable Operations Dashboard

Team Name:

[Team Name as registered on portal] • Team ID: [fill from portal]

Submission check: 5 slides • points, no paragraphs • export to PDF for portal upload

Idea / Proposed Solution

Describe your Idea / Solution / Prototype here • how it addresses the problem • innovation

PROBLEM IN ONE LINE

NE logistics fleets get stuck or crash on unseen hazards (landslides, floods, blackspots) — standard apps optimise only for speed.

- **Risk-aware A* engine:** custom cost $c(e) = t(e) \times [1 + \alpha \cdot r(e)]$ over OSMnx road graph; returns fastest + safest route side-by-side with time vs risk trade-off.
- **Risk model r(e):** 0.5 accident history + 0.3 surface condition + 0.2 weather exposure; missing evidence stays ‘unknown’, never silently safe.
- **Explainable output:** route + extra minutes + avoided exposure + evidence list operator can act on; single-access blockage raises air-drop / escalation alert.
- **Innovation:** safety inside the edge cost (not a post-filter); operator-tunable α dial keeps A* provably optimal per setting; verified vs Dijkstra on 300 cases.

A vs DIJKSTRA CONCEPT (SCHEMATIC, NO IMAGE)

FASTEST ROUTE (Dijkstra / standard)
Shorter • crosses hazard zone • lower time, higher exposure

▼ vs

RISK-AWARE ROUTE (RaahSetu A*)
Safer • avoids blackspots + weather • quantified extra minutes

Demo corridor: 75% lower modelled exposure for +7.7 min.
Prototype live on real OSM graph (5,814 nodes).

Technical Architecture & Technology Stack

Describe your Technology stack here • one explainable path from request to route decision

Fleet operator
request + α

→

FastAPI (Py 3.12)
request layer

→

OSMnx graph + PostGIS
hazard joins

→

Risk-aware A*
route compare

→

Ops dashboard
explain + act

PostgreSQL + PostGIS (hazards, reviews, fleet) • versioned scenario store • Open-Meteo live weather (keyless, free)

Frontend	Backend	Database	Deployment
React, TypeScript (NO screenshots in this deck)	Python 3.12, FastAPI OSMnx graph	Supabase PostgreSQL + PostGIS	Docker Kubernetes manifests

Why this wins: 44 automated tests • 300 randomised A* vs Dijkstra oracle checks • 14/14 live Supabase E2E steps • 8-state OSM extracts ready (273,494 ways) • 36 MB / 500 MB free tier

Use Cases

Describe your Use Cases here • who uses it and what changes on the ground

01 EMERGENCY RESPONSE

NDRF / SDRF fleets

Relief run around flooded / blocked / landslide-prone segments during extreme weather.

Outcome: on-time relief, fewer stranded consignments.

02 ESSENTIAL LOGISTICS

Heavy commercial trucks

Compare time vs risk before moving medicines, food, material on NE corridors.

Outcome: quantified detour cost before dispatch.

03 FIELD REPORTING LOOP

District officials

Geo-tagged closure/photo queued offline; reviewer accepts before it changes route cost.

Outcome: every closure has provenance, no rumours.

Operational loop: Request → Compare (fastest vs safest) → Explain (evidence) → Act | Special case — Risk Mitigation Mode: if single-access route is blocked, raise accessibility gap + air-drop escalation instead of a misleading route.

Dependencies & Show Stoppers

Describe your Dependencies / Show stopper here • maturity check: honest risks + shipped mitigations

DEPENDENCIES (WHAT WE NEED)

- **OSM / OSMnx road graph:** open network for all 8 NE states; versioned snapshots.
- **Hazard evidence:** reviewed landslide + accident-blackspot records (MoRTH / eDAR as scaling path).
- **Weather updates:** Open-Meteo live feed (free, keyless) mapped to routing scenarios.
- **Supabase + PostGIS:** project with spatial schema, RLS, evidence storage.

SHOW STOPPERS → RESPONSE (RISK | MITIGATION)

- **Paid traffic APIs:** bypassed — open OSMnx graph + custom A*; freshness shown explicitly.
- **Unverified coordinates:** never auto-applied; review queue (accept / reject) before route cost changes.
- **Single road blocked:** accessibility alert + last feasible plan + escalation path.
- **Low-connectivity fields:** PWA offline shell + IndexedDB queue with auto-retry on sync.